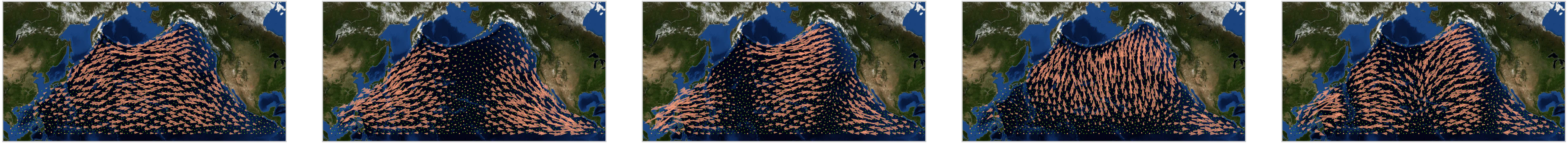
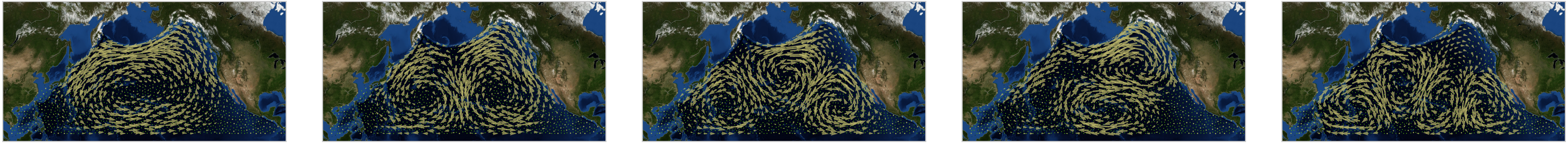
Continuous visualization

- Down Laplacian, its nonzero eigenspace spans the gradient space



λ_G, more divergent

- Up Laplacian, its nonzero eigenspace spans the curl space

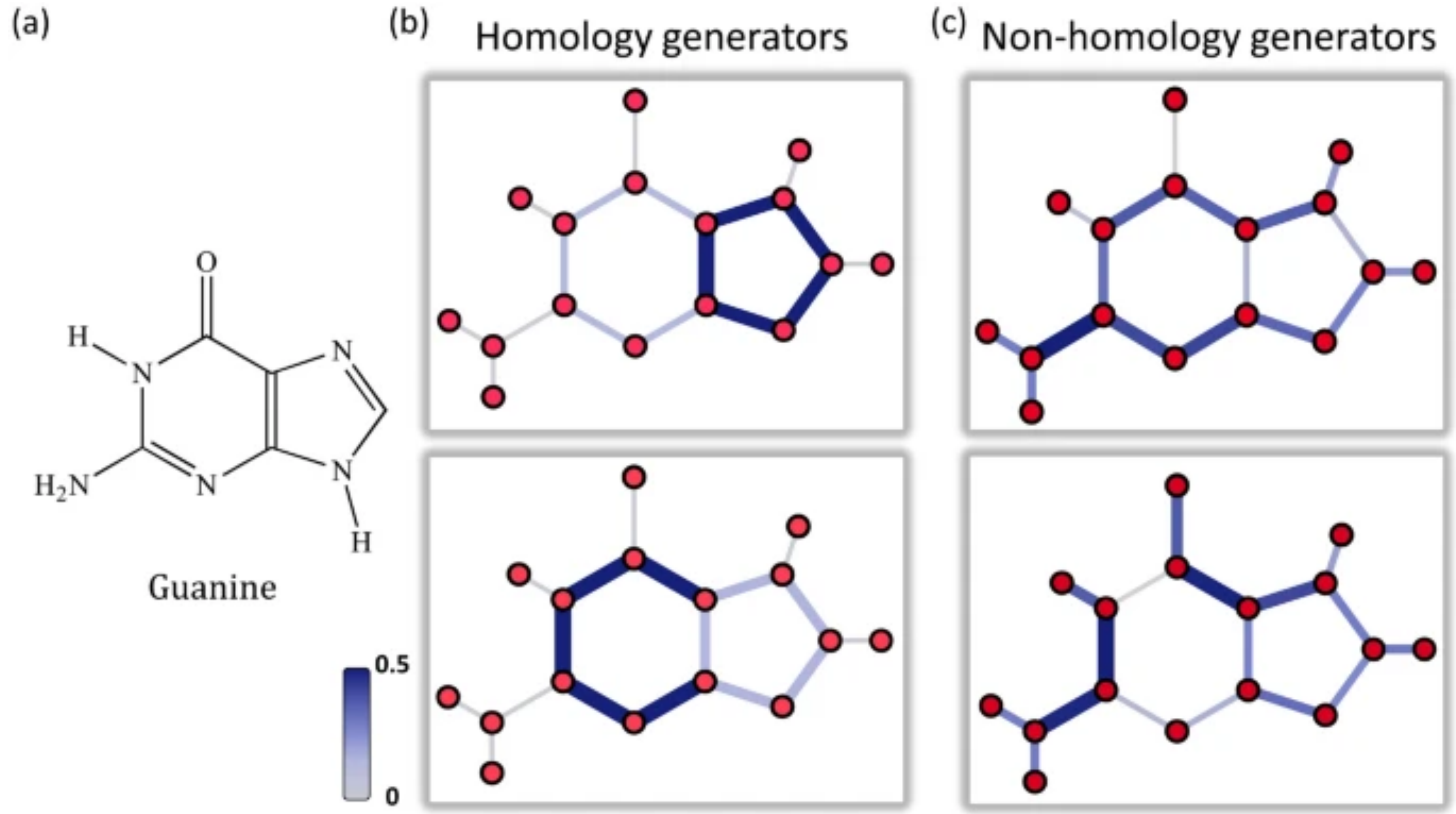


λ_C, more rotational

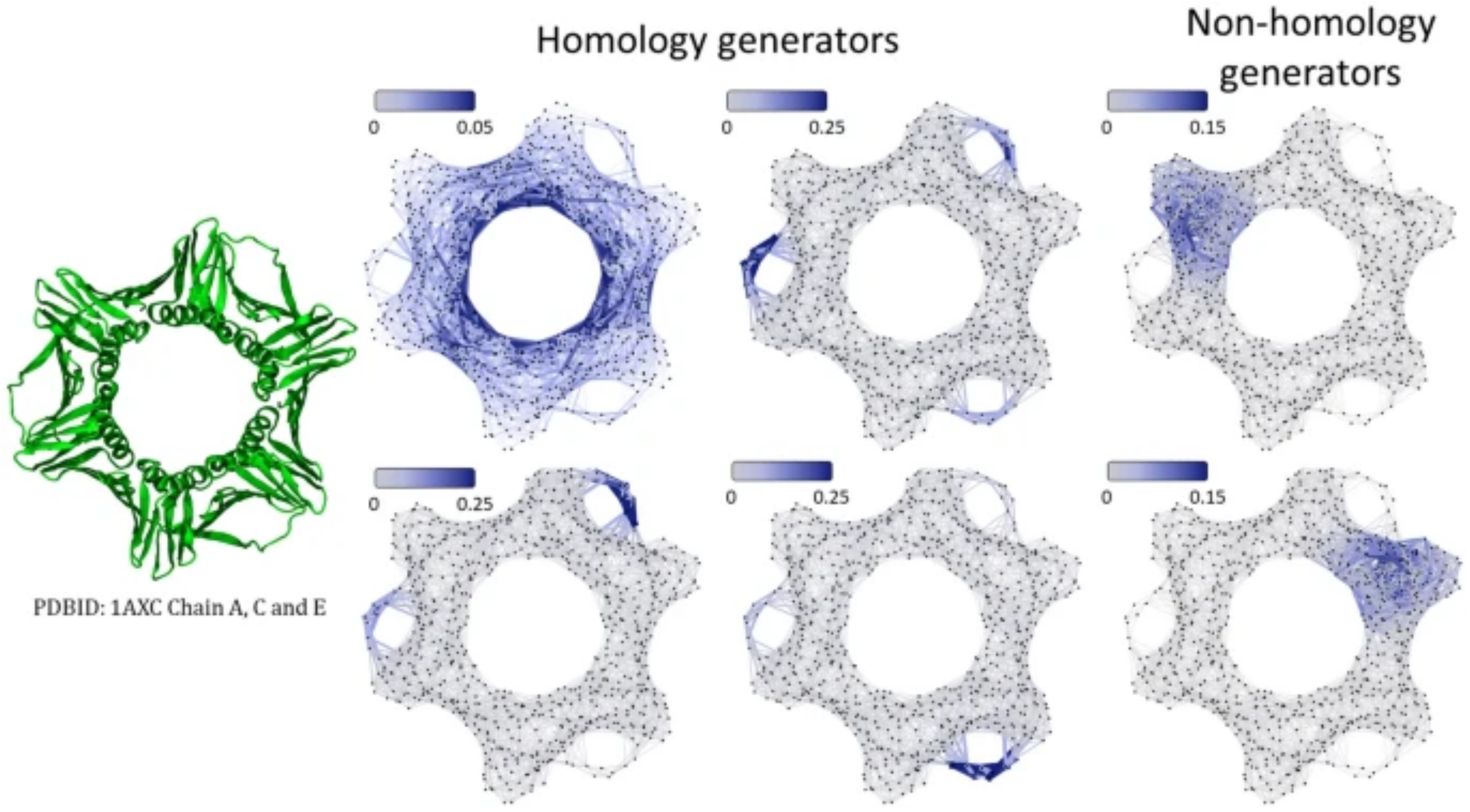
Chen, Yu-Chia et al. (2021) "Helmholtzian Eigenmap."

Applications in molecular data

Wei, R.K.J., Wee, J., Laurent, V.E. *et al.* Hodge theory-based biomolecular data analysis. *Sci Rep* (2022)



(a) The graph representation for Guanine. (b) The zero-eigenvalue-related eigenvectors are homology generators. Geometrically, their largest absolute values indicate the associated loop structures. (c) Nonzero-eigenvalue-related eigenvectors are more related to domain, cluster and community structures.



protein structure analysis for protein (ID:1AXC). Four 1D homology generators are from four zero eigenvalue related eigenvectors of 1D HL matrix. Two non homology generators are from two smallest nonzero eigenvalue related eigenvectors. Homology generators characterize loop and cycle structures, while non-homology generators indicate information about domains and communities.